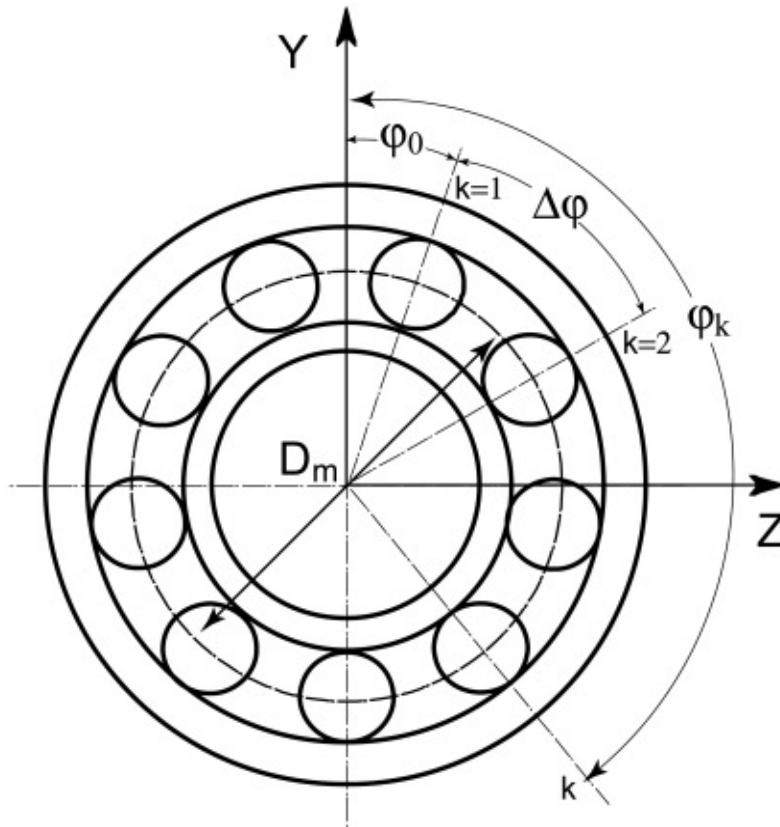




**CLEVELAND STATE
UNIVERSITY**

Class: CSU MCE#567 Final Project

Title: Design of Bearings of Various Lubrication Regimes

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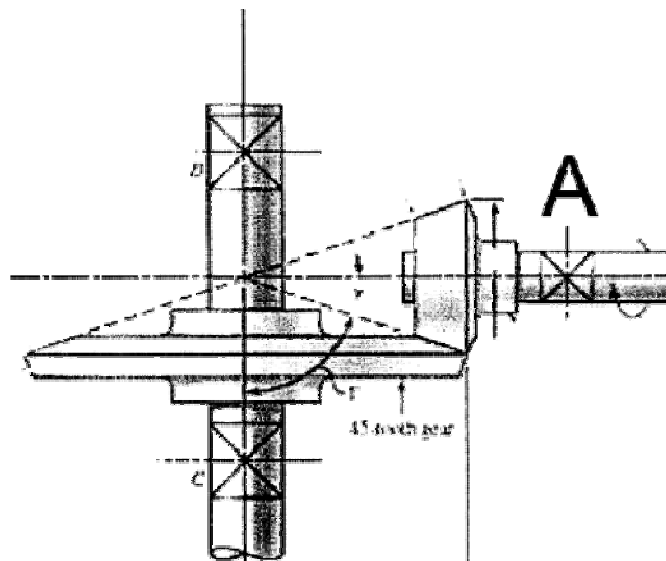
2. Problem#1: Hydrodynamic & Elastohydrodynamic Bearing Design:

Design appropriate bearings at point A. The load and speed parameters at point A are the following:

- Given this information, design the following bearing(s):

- Radial Journal
- Thrust

- Radial Roller
- Thrust



2.1.1. Radial Journal Bearing Design(Hydrodynamic Lubrication):

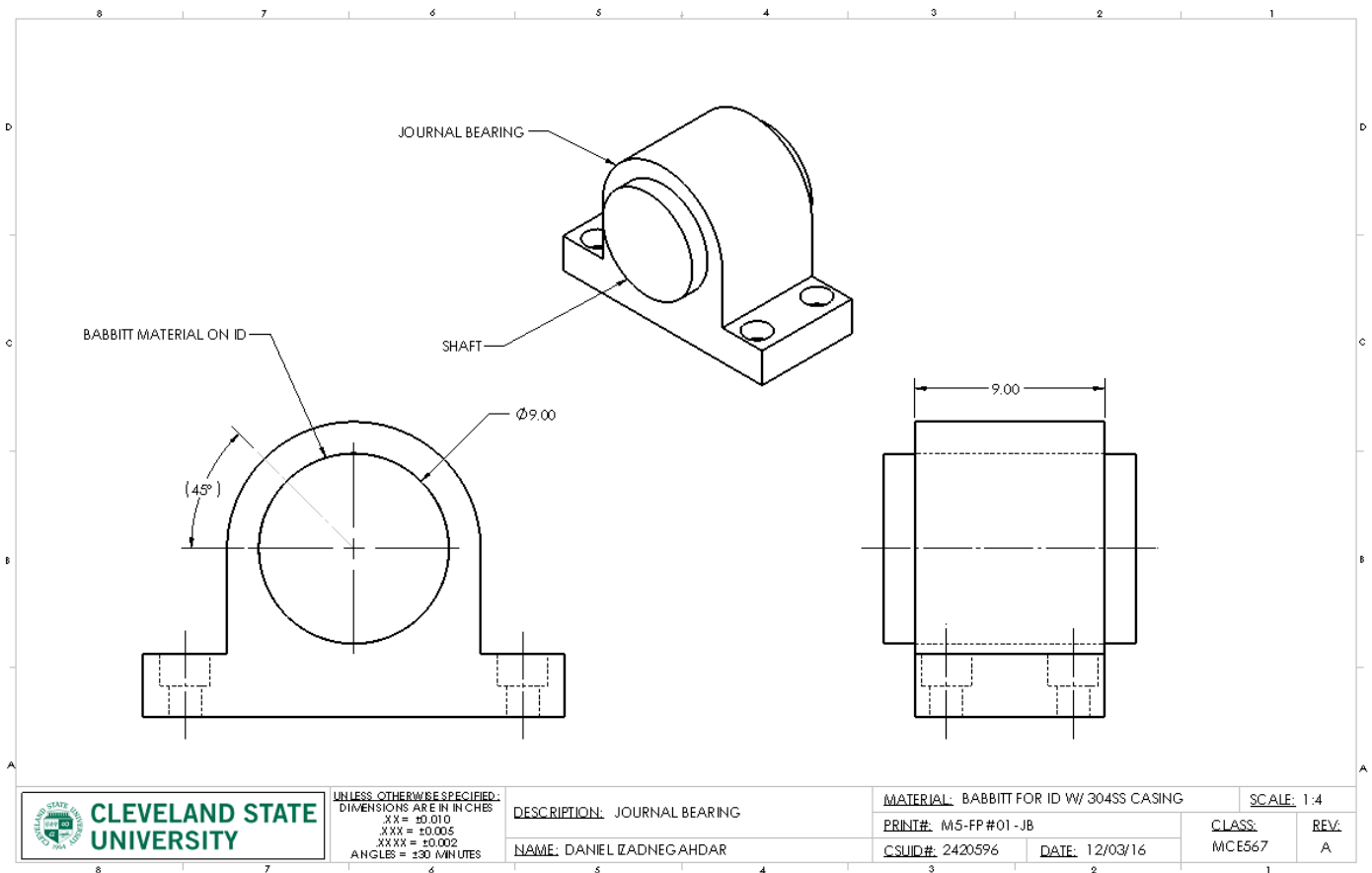
2.1.1.1. Calculations:

Journal Bearing Design Calculations							
Description	Symbol	Equation / Condition	SI Units		English Units		Comments
			Value	Unit	Value	Unit	
Given:							
Lubrication Regime	N/A	Given	Hydrodynamic	N/A	Hydrodynamic	N/A	None
Operating Condition	N/A	Given	Low Load/ High Speed	N/A	Low Load/ High Speed	N/A	None
Radial/Static Load	W(Radial)	Given	9000.00	N	2023.28	lbf	None
Rotational Speed(RPM)	Na(RPM)	Given	3600.00	RPM	3600.00	RPM	None
Bearing Supported Mass	m	Given	40.000	kg	88.185	lbf	None
Assume:							
Shaft Journal Radius	r	Assumed + Optimized	0.114	m	4.500	in	Low load assumes lower radius
Length to Diameter ratio	l/d	Assumed + Optimized	1.000	N/A	1.000	N/A	None
Lubricant	Lube	Assumed + Optimized	SAE20	N/A	SAE20	N/A	Higher speed = Lubricant with lower viscocity
Temperature inlet	Tin	Assumed + Optimized	82.229	°C	180.000	°F	None
Temperature Rise (Guessed)	ΔT1	Assumed + Optimized	2.778	°C	37.00	°F	This value has been iterated until ΔT1* is ± 5°F
Geometrical Calculations:							
Rotational Speed(RPS)	Na(RPS)	=Na/60	60.000	RPS	60.000	RPS	None
Shaft Journal Diameter	d	=2*r	0.229	m	9.000	in	None
Clearance	c	=r/1000	1.14E-04	m	0.005	in	None
Bearing Length	l	=(d)*(l/d)	0.23	m	9.000	in	None
Temperature (operating)	Top	=Tin+(ΔT1/2)	92.507	°C	198.500	°F	None
Viscosity	μ	=(FIG 12-12 @ T _{opt} @ Lube) OR: use derived equation(English Units Only): =88338 * T _{opt} ^ -2.63 @SAE10 =4.90E+06 * T _{opt} ^ -2.91 @SAE20 =2.00E+07 * T _{opt} ^ -3.11 @SAE30 =6.00E+07 * T _{opt} ^ -3.28 @SAE40 =4.00E+08 * T _{opt} ^ -3.52 @SAE50 =9.00E+08 * T _{opt} ^ -3.67 @SAE60 =2.00E+09 * T _{opt} ^ -3.80 @SAE70	0.007	Pa*sec	1.01E-06	psi-sec	See Appendix for Derivations
Nominal Bearing Pressure	P	=W/(d*l)	1.72E+05	Pa	24.979	psi	None
Sommerfield Number	S	=((μ*Na(RPS))/P)*(r/c)^2	2.423	N/A	2.423	N/A	None
Temperature Rise Dimensionless Variable	RHS	=(FIG 12-24 @ l/d @ S) OR: use derived equation: =(l/d = 1) 0.349109 + 6.00940*S + 0.07467*S^2 =(l/d = 1/2) 0.394552 + 6.392527*S - 0.036013*S^2 =(l/d = 1/4) 0.933828 +6.437512*S - 0.011048*S^2	15.19	N/A	15.19	N/A	See Appendix for Derivations
Temperature Rise (Calculated)	ΔT1*	=(RHS * P) / 9.7	3.949	°C	39.11	°F	None
Dimensionless Minimum Film Thickness Variable	ho*	= ho / c OR: (FIG 12-16 @ l/d @ S) OR: use piece-wise linear regression Appendix equations. OR: use derived equation(s): = c* (0.85*S ^ 0.09) @ l/d = 1.00 = c* (0.15ln(x) + 0.64) @ l/d = 0.50 = c* (0.20ln(x) + 0.39) @ l/d = 0.25	0.938	N/A	0.938	N/A	See Appendix for Derivations
Minimum Film Thickness	ho	=c * ho*	0.000	m	0.004	in	None

Dimensionless Coefficient of Friction Variable	f*	= f * (r/c) OR: (FIG 12-18 @ l/d @ S) OR: use piece-wise linear regression Appendix equations.	44.242	N/A	44.242	N/A	See Appendix for Derivations
Coefficient of Friction	f	=(c/r) * f*	0.044	N/A	0.044	N/A	None
Dimensionless Circumferential Flow Variable	Q*	= Q / (r * c * N * l) OR: (FIG 12-19 @ l/d @ S) OR: use piece-wise linear regression Appendix equations.	3.239	N/A	3.239	N/A	See Appendix for Derivations
Circumferential Flow	Q	= (R*C*Na*I) * Q*	0.001	m^3/s	35.42	in^3/s	None
Flow Ratio	Qratio	=Qs/Q OR: (FIG 12-19 @ l/d @ S) OR: use piece-wise linear regression Appendix equations.	0.087	N/A	0.087	N/A	See Appendix for Derivations
Side Flow	Qs	=(Q)* Qratio	5.07E-05	m^3/s	3.09	in^3/s	None
Stability Calculations:							
Damping constant (xx direction)	Cxx	=(FIG B-3 @ l/d @ S) OR: use piece-wise linear regression Appendix equations.	36.34	N/A	36.34	N/A	See Appendix for Derivations
Damping constant (yy direction)	Cyy	=(FIG B-3 @ l/d @ S) OR: use piece-wise linear regression Appendix equations.	2.20	N/A	2.20	N/A	See Appendix for Derivations
Spring constant (yy direction)	Kyy	=(FIG B-3 @ l/d @ S) OR: use piece-wise linear regression Appendix equations.	36.34	N/A	36.34	N/A	See Appendix for Derivations
Spring constant (xx direction)	Kxx	=(FIG B-3 @ l/d @ S) OR: use piece-wise linear regression Appendix equations.	1.02	N/A	1.02	N/A	See Appendix for Derivations
Damping constant (yx direction)	Cyx	=(FIG B-4 @ l/d @ S) OR: use piece-wise linear regression Appendix equations.	2.44	N/A	2.44	N/A	See Appendix for Derivations
Damping constant (xy direction)	Cxy	=(FIG B-4 @ l/d @ S) OR: use piece-wise linear regression Appendix equations.	1.24	N/A	1.24	N/A	See Appendix for Derivations
Spring constant (yx direction)	Kyx	=(FIG B-4 @ l/d @ S) OR: use piece-wise linear regression Appendix equations.	18.17	N/A	18.17	N/A	See Appendix for Derivations
Spring constant (yx direction)	Kxy	=(FIG B-4 @ l/d @ S) OR: use piece-wise linear regression Appendix equations.	31.86	N/A	31.86	N/A	See Appendix for Derivations
Mass constant	m	=(m*c*(2*PI()*(Na(RPS))^2)/W(Radial)	0.07	(SI)	27.87	(Eng)	None
Eigenvalue 11	α11	=EigenValues[-inv(M)*K -inv(M)*C; [1 0; 0 1] [0 0; 0 0]];	-1.5348 + 0.0000i	N/A	-1.5348 + 0.0000i	N/A	See eigenvalue matrices
Eigenvalue 12	α12	=EigenValues[-inv(M)*K -inv(M)*C; [1 0; 0 1] [0 0; 0 0]];	0.1292 + 0.9917i	N/A	0.1292 + 0.9917i	N/A	See eigenvalue matrices
Eigenvalue 21	α21	=EigenValues[-inv(M)*K -inv(M)*C; [1 0; 0 1] [0 0; 0 0]];	0.1292 - 0.9917i	N/A	0.1292 - 0.9917i	N/A	See eigenvalue matrices
Eigenvalue 22	α22	=EigenValues[-inv(M)*K -inv(M)*C; [1 0; 0 1] [0 0; 0 0]];	-0.0646 + 0.0000i	N/A	-0.0646 + 0.0000i	N/A	See eigenvalue matrices
Design Verifications:							
Temperature Rise Optimization	ΔT	-50°F < ΔT < 50°F (PASS) else (FAIL)	3.363	°C	38.05	°F	PASS
Temperature Rise Range ± 5°F	ΔT1* - ΔT1	-10°F < (ΔT1* - ΔT1) < 10°F(PASS) else (FAIL)	-16.608	°C	2.11	°F	PASS
Clearance Optimization	Cop	c = r/1000	0.0001	m	0.001	in	PASS
Stability	Stability	Negative Eigenvalues = Stable Positive Eigenvalues = Unstable	50%	N/A	50%	N/A	2 of the 4 eigenvalues are positive and at risk of being unstable

Matrix Variables				Matrix Values												
M Matrix	m		0		27.87		0									
	0		m		0		27.875									
K Matrix	Kxx		Kxy		1.02		31.86									
	Kyx		Kyy		18.17		36.34									
C Matrix	Cxx		Cxy		36.34		1.24									
	Cyx		Cyy		2.44		2.20									
I Matrix	-inv(M)*K		-inv(M)*C		-0.037		-1.143		-1.304		-0.044					
					-0.652		-1.304		-0.088		-0.079					
	1		0		0		0		1.000		0.000		0.000		0.000	
	0		1		0		0		0.000		1.000		0.000		0.000	
NF Matr	α_{11}		Excel EigenValues													
	α_{12}		Not a symmetric matrix													
	α_{21}		Not a symmetric matrix													
	α_{22}		Not a symmetric matrix													
NF Matr	α_{11}		Matlab EigenValues													
	α_{12}		-1.5348 + 0.0000i													
	α_{21}		0.1292 + 0.9917i													
	α_{22}		0.1292 - 0.9917i													
		-0.0646 + 0.0000i														

2.1.1.2. General Drawing:



2.1.1.3. Results Summary: For this *hydrodynamic journal bearing*, the recommended bearing parameters are $r = 4.50$ ", *Length-to-Diameter* ratio of 1.0", a clearance of 0.005", and SAE20 for the lubricant. The SAE20 lubricant is selected because of its lower viscosity to handle the relatively high bearing speed. This bearing provides a temperature rise of 38 ± 5 °F, a minimum film thickness of 0.004", a friction coefficient of 0.044, a circumferential flow rate of 35.42 in³/s, and a side flow rate of 3.09 in³/s. This design is favorable because the temperature rise is less than 50 °F but the stability may be a concern since 2 out of the 4 eigenvalues are positive (all 4 must be negative) and the Sommerfeld number is not less than 0.2. To improve stability, the Elastohydrodynamic roller bearing designed in section 1.1.3 is a better design for stability because EHL bearings are typically more stable than journal bearings.

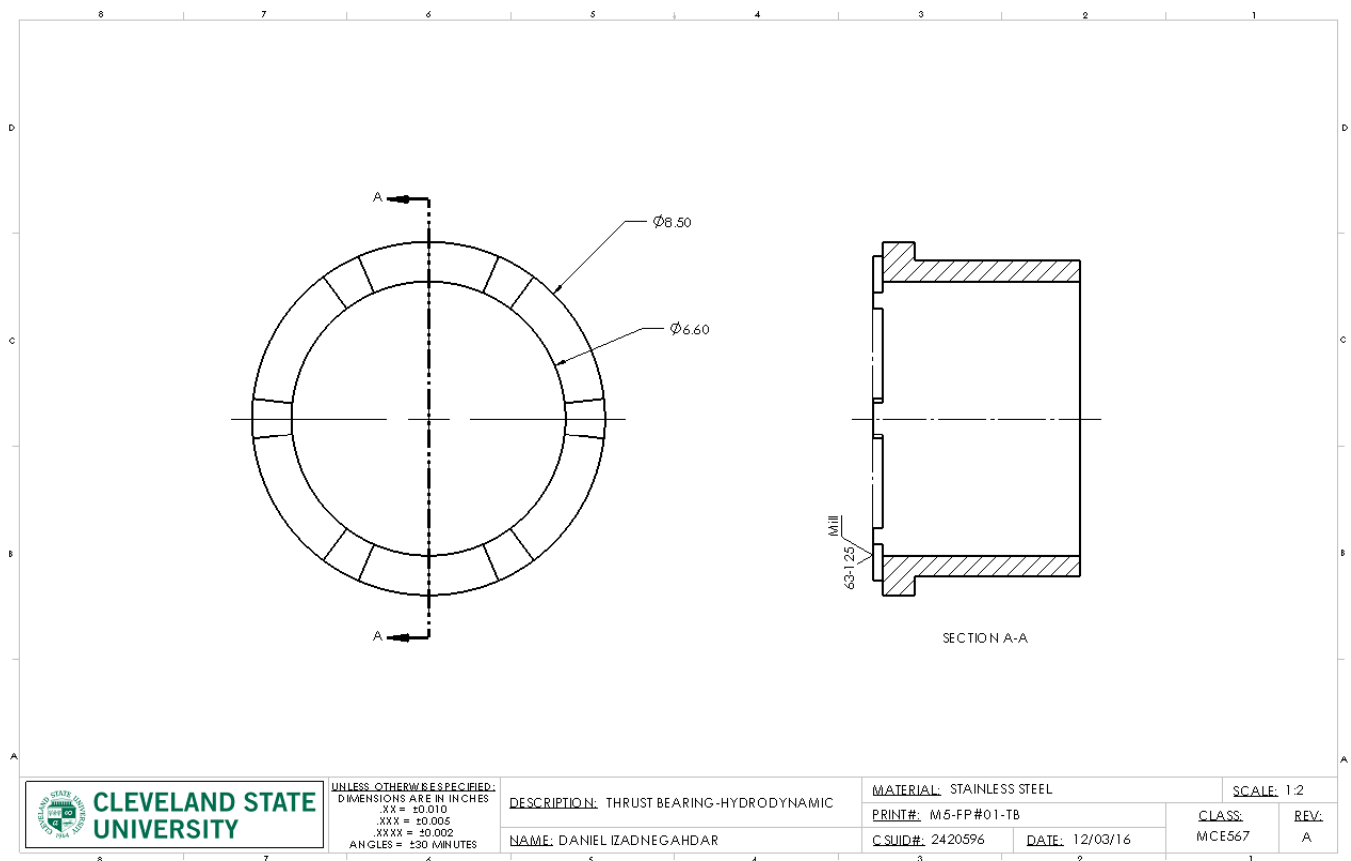
2.1.2. Thrust Bearing Design (Hydrodynamic Lubrication):

2.1.2.1. Calculations:

Thrust Bearing Design Calculations							
Description	Symbol	Equation / Condition	SI Units		English Units		Comments
			Value	Unit	Value	Unit	
Given:							
Lubrication Regime	N/A	Given	Hydrodynamic	N/A	Hydrodynamic	N/A	None
Operating Condition	N/A	Given	Low Load/ High Speed	N/A	Low Load/ High Speed	N/A	None
Type of Bearing	N/A	Given	Fixed	N/A	Fixed	N/A	None
Normal Load	W(Thrust)	Given	2000.00	N	449.62	lbf	None
Rotational Speed (RPM)	Na(RPM)	Given	3600.00	RPM	3600.00	RPM	None
Assume:							
Operating Temperature	T _{opt}	Assumed + Optimized	82.23	°C	180.00	°F	None
# of Pads	No	Assumed + Optimized	6.00	N/A	6.00	N/A	None
Lubricant	Lube	Assumed + Optimized	SAE20	N/A	SAE20	N/A	Higher speed = Lubricant with lower viscosity
Length-to-Width Ratio	λ	Assumed + Optimized	1.00	N/A	1.00	N/A	None
Outer Radius	r _o	Assumed + Optimized	0.11	m	4.25	in	None
Inner Radius	r _i	Assumed + Optimized	0.08	m	3.30	in	None
Geometrical Calculations (FIXED bearing) :							
Rotational Speed (RPS)	Na(RPS)	=Na/60	60.00	RPS	60.00	RPS	None
Dimensionless Film Thickness(fixed)	Ho	=(FIG 9-6 @ Max-Normal @ λ) OR: Derived Equation: =0.001*(λ)^5 - 0.019*(λ)^4 + 0.101*(λ)^3 - 0.201*(λ)^2 + 0.014*(λ) + 0.846	0.74	N/A	0.74	N/A	None
Dimensionless Normal Load(fixed)	W(fixed)	=6*LN((Ho+1)/Ho)-(12)/(1+2*Ho)	0.29	N/A	0.29	N/A	None
Absolute Viscosity	Eta(η)	=(FIG 12-12 @ T _{opt} @ Lube) OR: use derived Equation(English Units Only): =SAE(10) 88338 * T _{opt} ^ -2.63 =SAE(20) 4.90E+06 * T _{opt} ^ -2.91 =SAE(30) 2.00E+07 * T _{opt} ^ -3.11 =SAE(40) 6.00E+07 * T _{opt} ^ -3.28 =SAE(50) 3.00E+08 * T _{opt} ^ -3.52 =SAE(60) 9.00E+08 * T _{opt} ^ -3.67 =SAE(70) 2.00E+09 * T _{opt} ^ -3.80	9.24E-03	Pa*sec	1.34E-06	psi-sec	None
Rotational Speed(rads)	w	=(Na(RPS))*2*PI()	376.99	rad/s	376.99	rad/s	None
Surface velocity	U _b	=0.5*(r _o +r _i)*w	36.15	m/s	1423.14	in/sec	None
Pad Width	b	=r _o - r _i	2.41E-02	m	0.95	in	None
Pad Length	l	=λ * b	2.41E-02	m	0.95	in	None
Load per Pad	Wz(pad)	=W(Thrust) / (# of pads)	333.33	N	74.94	lbf	None
Minimum Film Thickness(fixed)	h _o	=(Ho*I)*SQRT(((W(fixed)*Eta*U _b *b)/(Wz(pad))))	4.74E-05	m	0.0019	in	None
Shoulder Height	Sh	=h _o /Ho	6.39E-05	m	0.0025	in	None
Surface Finish	Ra	=(Table 9.2 @ h _o) OR: If Statement: '=IF(G26<0.0001, "4-8 Ra[Superfinish]", IF(G26<0.00025, "8-16 Ra[Lap]", IF(G26<0.0005, "16-32 Ra[Grind]", IF(G26<0.001, "32-63 Ra[Mill]", IF(G26<0.002, "63-125 Ra[Shape]", "No finish required")))))	63-125 Ra[Mill]	Ra(μm)	63-125 Ra[Mill]	Ra(μin)	None
Design Verifications:							
Polishing expense	N/A	h _o > 0.0002in	4.74E-05	m	0.0019	in	PASS
Stability (Avoid flooding)	N/A	h _o < 0.003in	4.74E-05	m	0.0019	in	PASS

Geometrical Calculations (PIVOT bearing) (NOT USED FOR DESIGN):							
Type of Bearing	N/A	N/A	Pivot	N/A	Pivot	N/A	None
Pivot location from inlet	ns	=(Figure 9.12 @λ @Max Normal Load)	1.52E-02	m	0.60	in	Manually inspected
Dimensionless Load Function	Wz*	= √Wz OR: (Figure 9.13 @λ @ns)	0.26	N/A	0.26	N/A	Manually inspected
Dimensionless Normal Load	Wz(pivot)	= Wz* ^ 2	0.068	N/A	0.068	N/A	None
Dimensionless Minimum Film Thickness	Ho*	=√Wz / Ho OR: (Figure 9.15a @λ @ns)	0.300	N/A	0.300	N/A	Manually inspected
Dimensionless Film Thickness	Ho	'=[(Figure 9.15 @ λ @ns)] * √(Wz)	0.87	N/A	0.87	N/A	None
Minimum Film Thickness	ho	'=(Ho*I)*SQRT((Wz*Eta*Ub*b)/(Wz(pivot)))	2.674E-05	m	0.0011	in	None
Surface Finish	Ra	=(Table 9.2 @ ho) OR: If Statement: '=IF(G26<0.0001, "4-8 Ra[Superfinish]", IF(G26<0.00025, "8-16 Ra[Lap]", IF(G26<0.0005, "16-32 Ra[Grind]", IF(G26<0.001, "32-63 Ra[Mill]", IF(G26<0.002, "63-125 Ra[Shape]", "No finish required")))))	63-125 Ra[Mill]	Ra(μm)	63-125 Ra[Mill]	Ra(μin)	None

2.1.2.2. General Drawing:



2.1.2.3. Results Summary: For this *hydrodynamic fixed thrust* bearing, the recommended parameters are $r_o = 4.25''$, $r_i = 3.30''$, 6 pads, *Length-to-Width* ratio of 1.00, and SAE20 for the recommended lubricant. The SAE20 lubricant is selected because of its lower viscosity to handle the relatively high bearing speed. This bearing provides a minimum film thickness of 0.0019" which isn't too small to require an expensive lapping polish nor too large to cause unstable flooding. The geometric dimensions were optimized to keep this film thickness within that range.

2.1.3. Radial Roller Bearing Design (Elastohydrodynamic Regime):

2.1.3.1. L10 Life Calculations:

Elastohydrodynamic Regime Bearing Design Calculations (RADIAL ROLLER)							
Description	Symbol	Equation / Condition	SI Units		English Units		Comments
			Value	Unit	Value	Unit	
Given (Single Bearing):							
Lubrication Regime	N/A	Given	Hydrodynamic	N/A	Hydrodynamic	N/A	None
Operating Condition	N/A	Given	Low Load/ High Speed	N/A	Low Load/ High Speed	N/A	None
Bearing Load Style	Bls	Given	Radial	N/A	Radial	N/A	OPTIONS: Radial/Thrust (Select Radial for Angular)
Bearing Hardware Style	Bhs	Given	Roller	N/A	Roller	N/A	OPTIONS: Ball/Roller
Bearing Special	Bs	Given	None	N/A	None	N/A	OPTIONS: None/Filling/Groove/Self-Aligning/Separable/Cylindrical/Needle/Spherical
Thrust Load	Fa	Given	2000.00	N	449.62	lbf	None
Radial Load	Fr	Given	9000.00	N	2023.28	lbf	None
Rotational Speed(RPM)	Na(RPM)	Given	3600.00	RPM	3600.00	RPM	None
Assume(Single Bearing):							
# of Rows of Rolling Element	i	Assumed + Optimized	1.00	N/A	1.00	N/A	OPTIONS: Single(1) / Double (2)
# of Rolling Elements per Row	Z	Assumed + Optimized	15.00	Balls/ Rollers	15.00	Balls/ Rollers	Optimized for L10
Outer Diameter	dout	Assumed + Optimized	0.20	m	8.00	in	Optimized for L10
Bore Diameter	din	Assumed + Optimized	0.15	m	6.00	in	Optimized for L10
Bearing Width	dw	Assumed + Optimized	0.04	m	1.50	in	None
Rolling Element Diameter	d	Assumed + Optimized	1.27E-02	m	0.50	in	Optimized for L10
Nominal Contact Angle	α	Assumed + Optimized	50.00	deg	50.00	deg	0/5/10/15/20/25/30/35/40/4
Roller Length	l	=IF(Bhs = "Ball", "N/A", (IF(Bhs = "Roller", "Value Required")))	1.00	m	1.00	in	For roller, length value required!
Ball Diameter Limit (11-28-16 Handout)	dlim	Assumed + Optimized	0.0250	m	0.984	in	None
Calculations (Single Bearing):							
Load-Life Exponent	p	=IF(Bhs="Ball", 3, (IF(Bhs="Roller", 10/3)))	3.33	N/A	3.33	N/A	None
Pitch Diameter	dm	=AVERAGE(dout, din)	0.18	m	7.00	in	None
Table 1.1 Fcm Coefficient	Tc-fcm	=(d*COS(RADIANS(α)))/dm	0.05	N/A	0.0459	N/A	None

Material Coefficient	fcm	= (Table 1.1 - Table 1.6 @Bls @Bhs @i) OR: =IF(AND(Bls="Radial", Bhs="Ball", i = 1), FORECAST(Tc-fcm, OFFSET(TABLES!C11:C50, MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50, 1)-1, 0, 2), OFFSET(TABLES!\$B\$11:\$B\$50, MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50, 1)-1, 0, 2)), IF(AND(Bls="Radial", Bhs="Ball", i = 2), FORECAST(Tc-fcm, OFFSET(TABLES!D11:D50, MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50, 1)-1, 0, 2), OFFSET(TABLES!\$B\$11:\$B\$50, MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50, 1)-1, 0, 2)), IF(AND(Bls="Radial", Bhs="Roller"), FORECAST(Tc-fcm, OFFSET(TABLES!G11:G50, MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50, 1)-1, 0, 2), OFFSET(TABLES!\$B\$11:\$B\$50, MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50, 1)-1, 0, 2)), IF(AND(Bls="Thrust", Bhs="Ball", a = 45), FORECAST(Tc-fcm, OFFSET(TABLES!I11:I40, MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50, 1)-1, 0, 2), OFFSET(TABLES!\$B\$11:\$B\$50, MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50, 1)-1, 0, 2)), IF(AND(Bls="Thrust", Bhs="Ball", a = 60), FORECAST(Tc-fcm, OFFSET(TABLES!J11:J30, MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50, 1)-1, 0, 2), OFFSET(TABLES!\$B\$11:\$B\$50, MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50, 1)-1, 0, 2)), IF(AND(Bls="Thrust", Bhs="Ball", a = 75), FORECAST(Tc-fcm, OFFSET(TABLES!K11:K20, MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50, 1)-1, 0, 2), OFFSET(TABLES!\$B\$11:\$B\$50, MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50, 1)-1, 0, 2)), IF(AND(Bls="Thrust", Bhs="Roller", a = 50), FORECAST(Tc-fcm, OFFSET(TABLES!M11:M36, MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50, 1)-1, 0, 2), OFFSET(TABLES!\$B\$11:\$B\$50, MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50, 1)-1, 0, 2)), IF(AND(Bls="Thrust", Bhs="Roller", a = 65), FORECAST(Tc-fcm, OFFSET(TABLES!N11:N30, MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50, 1)-1, 0, 2), OFFSET(TABLES!\$B\$11:\$B\$50, MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50, 1)-1, 0, 2))	7186.28	N/A	7186.28	N/A	Integrated: Radial Ball Radial Roller Thrust Ball (45,60,75 ONLY) Thrust Roller (50,65,80 ONLY)
Basic Dynamic Load Capacity	C	=(Eq. 1.5 - 1.11 @Bls @Bhs @d @a) OR: =IF(AND(Bls="Radial", Bhs = "Ball", d<=dlim), (fcm*((I)*COS(RADIANS(a)))^0.7)*(Z^(2/3))*(d^1.8)), IF(AND(Bls="Radial", Bhs = "Ball", d>dlim), (fcm*((I)*COS(RADIANS(a)))^0.7)*(Z^(2/3))*(d^1.4)), IF(AND(Bls="Radial", Bhs = "Roller"), (fcm*((I)*COS(RADIANS(a)))^(7/9))*(Z^(3/4))*(d^(29/27)))), IF(AND(Bls="Thrust", Bhs = "Ball", α<90), (fcm*((I)*COS(RADIANS(a)))^0.7)*(TAN(RADIANS(a))*Z^(2/3))*(d^1.8))), IF(AND(Bls="Thrust", Bhs = "Roller", α<90), (fcm*((I)*COS(RADIANS(a)))^(7/9))*((TAN(RADIANS(a))*Z^(3/4))*(d^(29/27))))), IF(AND(Bls="Thrust", Bhs = "Ball", α=90), (fcm*((I)^0.7)*(Z^(2/3))*(d^1.8)), IF(AND(Bls="Thrust", Bhs = "Roller", α=90), (fcm*((I)^(7/9))*(Z^(3/4))*(d^(29/27))))))))))	82063.41	N	18448.60	lbf	Eq. 1.5-1.11 integrated
Load Ratio	Lr	=IF(Fr = 0, "1000", Fa/Fr)	0.22	N/A	0.22	N/A	If number is larger than 999, assume infinity
Relative Axial Load	RAL	=Fa/(l * Z * d^2)	0.83	N/(mm^2)	119.90	lb/in^2	None
Limit Value	e	=(Table 1.7 - 1.10 @α @Bls @Bhs) OR: use piece-wise linear regression equations.	0.00	N/A	0.00	N/A	Table 1.7 - 1.10 integrated
Dynamic Radial Load Factor	X	=(Table 1.7 - 1.10 @α @Bls @Bhs) OR: use piece-wise linear regression equations.	0.40	N/A	0.40	N/A	Table 1.7 - 1.10 integrated
Dynamic Axial Load Factor	Y	=(Table 1.7 - 1.10 @α @Bls @Bhs) OR: use piece-wise linear regression equations.	0.34	N/A	0.34	N/A	Table 1.7 - 1.10 integrated
Equivalent Load	P	=XFr + YFa	4271.28	N	960.22	lbf	Eq 1.12
L10 Life (in cycles)	L10(cycles)	=((C/P)^p)*10^6	1.90E+10	Cycles	1.90E+10	Cycles	Eq 1.4
L10 Life (in hours)	L10(hours)	=L10(Cycles) / (Na(RPM)*60min/hr)	87939.67	Hours	87939.67	Hours	None
L10 Life (in days)	L10(days)	=L10(Hours)/24	3664.15	Days	3664.15	Days	None
Design Verifications:							
Life optimization	None	L10(days) > 365 (AVERAGE(din, dout)*PI() - (d*Z)^2)/ (AVERAGE(din, dout)*PI()) > 0	3664.15	Days	3664.15	Days	PASS
Can #of balls of given diameter fit inside given raceways?	None	circumference of raceway pitch diameter > 2.00 * (ball diameter * # of balls) (NOTE: 2.00 is a safety clearance factor for inner spacing between balls for positioning mechanism)	None	%	32%	in	PASS

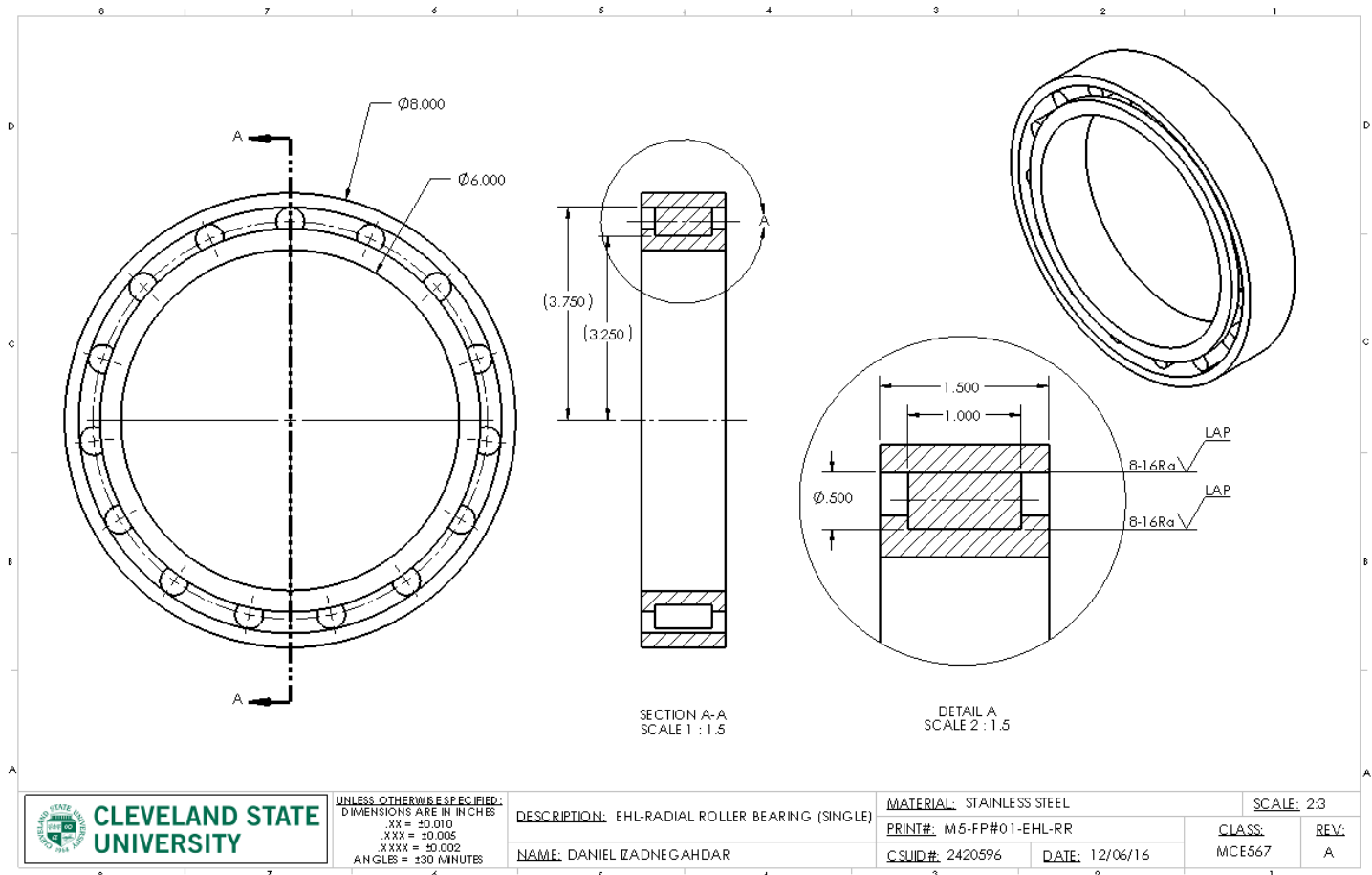
2.1.3.2. Film Thickness Calculations (Inner Race):

Elastohydrodynamic Regime Film Thickness Calculations (RADIAL ROLLER)							
Description	Symbol	Equation / Condition	SI Units		English Units		Comments
			Value	Unit	Value	Unit	
Given:							
Inner Race Diameter	di	Recycled	0.17	m	6.50	in	Calculated from given dimensions of L10 calculator
Outer Race Diameter	do	Recycled	0.19	m	7.50	in	Calculated from given dimensions of L10 calculator
Roller Diameter	d	Recycled	0.01	m	0.50	in	Recycled from L10 calculator
Roller Length	l	Recycled	0.03	m	1.00	in	Recycled from L10 calculator
Number of Rollers	n	Recycled	15.00	N/A	15.00	N/A	Recycled from L10 calculator
Radial Load	wz(total)	Recycled	9000.00	N	2023.28	lbf	Recycled from L10 calculator
Inner-Race Angular Velocity	wi	Given	376.99	rad/s	376.99	rad/s	Na(RPM) converted to rads/sec
Outer-Race Angular Velocity	wo	Given	0.00	rad/s	0.00	rad/s	Recycled from L10 calculator
Absolute Viscosity	η0	Recycled	0.01	Pascal-sec	1.34E-06	psi-sec	Property of lubricant specified
Modulus of Elasticity	E	Recycled	2.075E+11	N/m^2 (Pascal)	30095144.17	lbf/in^2	Property of material specified
Viscosity-Pressure Coefficient	ξ	Recycled	2.20E-08	m^3/N	0.0060	in^3/lbf	Property of material specified
Race Angular Velocity	v	Recycled	10.00	m/s	393.70	in/s	Property of material specified
Poisson's Ratio of both materials	ν	Recycled	0.30	N/A	0.30	N/A	Property of material specified
Assume:							
Solid B	b	Assumed + Optimized	INNER	N/A	INNER	N/A	OPTION: INNER/OUTER
Solid A	a	Assumed + Optimized	ROLLER	N/A	ROLLER	N/A	None
Calculations:							
Outer Race Curvature Radius (b solid, x direction)	rbx(outer)	= do/2 * -1	-0.10	m	-3.75	in	None
Inner Race Curvature Radius (b solid, x direction)	rbx(inner)	= di/2	0.08	m	3.25	in	None
Ball Curvature Radius (a solid, x direction)	rax(ball)	= d/2	0.01	m	0.25	in	None
Equivalent Radius (x direction)	Rx	= 1 / ((1/rax) + (1/rbx))	0.01	m	2.32E-01	in	None
Radial Load(ball)	wz	= 4/n * wz(total)	2400.00	N	539.54	lbf	None
Equivalent Modulus of Elasticity	E'	= E/ (1 - ν^2)	2.28E+11	N/m^2	3.31E+07	lbf/in^2	None
Dimensionless Load	W'	= wz/ (E' * Rx * l)	7.03E-05	N/A	7.03E-05	N/A	None
Dimensionless Speed	U	= (v * η0)/ (E' * Rx)	6.88E-11	N/A	6.88E-11	N/A	None
Dimensionless Material Property	G	= ξ * E'	1.97E+05	N/A	197498.03	N/A	None
Dimensionless Film Thickness	He,min	= hmin/Rx = 1.714 * (W'^-0.128) * (U^0.694) * (G^0.568)	5.26E-04	N/A	5.26E-04	N/A	None
Film Thickness (m)	hmin	= 'Rx * He,min	1.22E-04	m	1.22E-04	in	None
Surface Finish	Ra	=(Table 9.2 @ ho) OR: If Statement: '=IF(G26<0.0001, "4-8 Ra[Superfinish]", IF(G26<0.00025, "8-16 Ra[Lap]", IF(G26<0.0005, "16-32 Ra[Grind]", IF(G26<0.001, "32-63 Ra[Mill]", IF(G26<0.002, "63-125 Ra[Shape]", "No finish required")))))	8-16 Ra[Lap]	Ra(μm)	8-16 Ra[Lap]	Ra(μin)	None

2.1.3.3. Film Thickness Calculations (Outer Race):

Elastohydrodynamic Regime Film Thickness Calculations (RADIAL ROLLER)							
Description	Symbol	Equation / Condition	SI Units		English Units		Comments
			Value	Unit	Value	Unit	
Given:							
Inner Race Diameter	di	Recycled	0.17	m	6.50	in	Calculated from given dimensions of L10 calculator
Outer Race Diameter	do	Recycled	0.19	m	7.50	in	Calculated from given dimensions of L10 calculator
Roller Diameter	d	Recycled	0.01	m	0.50	in	Recycled from L10 calculator
Roller Length	l	Recycled	0.03	m	1.00	in	Recycled from L10 calculator
Number of Rollers	n	Recycled	15.00	N/A	15.00	N/A	Recycled from L10 calculator
Radial Load	wz(total)	Recycled	9000.00	N	2023.28	lbf	Recycled from L10 calculator
Inner-Race Angular Velocity	wi	Given	376.99	rad/s	376.99	rad/s	Na(RPM) converted to rads/sec
Outer-Race Angular Velocity	wo	Given	0.00	rad/s	0.00	rad/s	Recycled from L10 calculator
Absolute Viscosity	η0	Recycled	0.01	Pascal-sec	1.34E-06	psi-sec	Property of lubricant specified
Modulus of Elasticity	E	Recycled	2.075E+11	N/m^2 (Pascal)	30095144.17	lbf/in^2	Property of material specified
Viscosity-Pressure Coefficient	ξ	Recycled	2.20E-08	m^3/N	0.0060	in^3/lbf	Property of material specified
Race Angular Velocity	v	Recycled	10.00	m/s	393.70	in/s	Property of material specified
Poisson's Ratio of both materials	v(nu)	Recycled	0.30	N/A	0.30	N/A	Property of material specified
Assume:							
Solid B	b	Assumed + Optimized	OUTER	N/A	OUTER	N/A	OPTION: INNER/OUTER
Solid A	a	Assumed + Optimized	ROLLER	N/A	ROLLER	N/A	None
Calculations:							
Outer Race Curvature Radius (b solid, x direction)	rbx(outer)	= do/2 * -1	-0.10	m	-3.75	in	None
Inner Race Curvature Radius (b solid, x direction)	rbx(inner)	= di/2	0.08	m	3.25	in	None
Ball Curvature Radius (a solid, x direction)	rax(ball)	= d/2	0.01	m	0.25	in	None
Equivalent Radius (x direction)	Rx	= 1 / ((1/rax) + (1/rbx))	0.01	m	2.68E-01	in	None
Radial Load(ball)	wz	= 4/n * wz(total)	2400.00	N	539.54	lbf	None
Equivalent Modulus of Elasticity	E'	= E/ (1 - v^2)	2.28E+11	N/m^2	3.31E+07	lbf/in^2	None
Dimensionless Load	W'	= wz/ (E' * Rx * l)	6.09E-05	N/A	6.09E-05	N/A	None
Dimensionless Speed	U	= (v * η0)/ (E' * Rx)	5.96E-11	N/A	5.96E-11	N/A	None
Dimensionless Material Property	G	= ξ * E'	1.97E+05	N/A	197498.03	N/A	None
Dimensionless Film Thickness	He,min	= hmin/Rx = 1.714 * (W'^0.128) * (U^0.694) * (G^0.568)	4.85E-04	N/A	4.85E-04	N/A	None
Film Thickness (m)	hmin	= 'Rx * He,min	1.30E-04	m	1.30E-04	in	None
Surface Finish	Ra	=(Table 9.2 @ ho) OR: If Statement: '=IF(G26<0.0001, "4-8 Ra[Superfinish]", IF(G26<0.00025, "8-16 Ra[Lap]", IF(G26<0.0005, "16-32 Ra[Grind]", IF(G26<0.001, "32-63 Ra[Mill]", IF(G26<0.002, "63-125 Ra[Shape]", "No finish required")))))	8-16 Ra[Lap]	Ra(μm)	8-16 Ra[Lap]	Ra(μin)	None

2.1.3.4. General Drawing:



2.1.3.5. Results Summary: As opposed to journal bearings, Elastohydrodynamic bearings typically have a finite life. For this reason, the L10 Life was calculated as shown in section 1.1.3.1. It is estimated that after 3,665 days, 10% of the bearings will fail. The geometric dimensions were optimized to maximize this value.

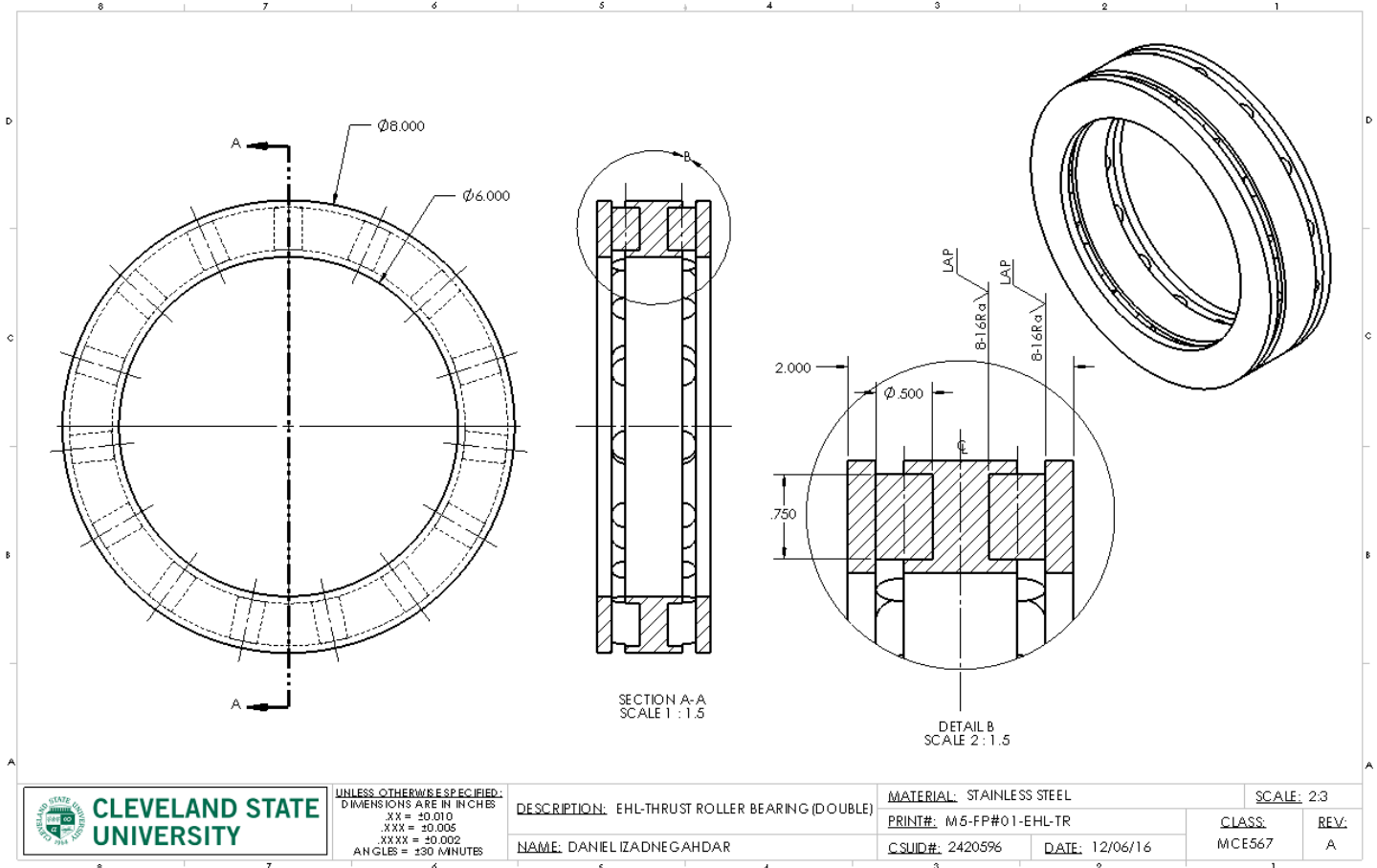
2.1.4. Thrust Bearing Design(Elastohydrodynamic Regime):

2.1.4.1. Calculations:

Elastohydrodynamic Regime Bearing Design Calculations (THRUST ROLLER)							
Description	Symbol	Equation / Condition	SI Units		English Units		Comments
			Value	Unit	Value	Unit	
Given (Single Bearing):							
Lubrication Reqime	N/A	Given	Hydrodynamic	N/A	Hydrodynamic	N/A	None
Operating Condition	N/A	Given	Low Load/ High Speed	N/A	Low Load/ High Speed	N/A	None
Bearing Load Style	Bls	Given	Thrust	N/A	Thrust	N/A	OPTIONS: Radial/Thrust (Select Radial for Angular)
Bearing Hardware Style	Bhs	Given	Roller	N/A	Roller	N/A	OPTIONS: Ball/Roller
Bearing Special	Bs	Given	None	N/A	None	N/A	OPTIONS: None/Filling/Groove/Self-Aligning/Separable/Cylindrical/Needle/Spherical
Thrust Load	Fa	Given	2000.00	N	449.62	lbf	None
Radial Load	Fr	Given	9000.00	N	2023.28	lbf	None
Rotational Speed(RPM)	Na(RPM)	Given	3600.00	RPM	3600.00	RPM	None
Assume(Single Bearing):							
# of Rows of Rolling Element	i	Assumed + Optimized	2.00	N/A	2.00	N/A	OPTIONS: Single(1) / Double (2)
# of Rolling Elements per Row	Z	Assumed + Optimized	15.00	Balls/ Rollers	15.00	Balls/ Rollers	Optimized for L10
Outer Diameter	dout	Assumed + Optimized	0.20	m	8.00	in	Optimized for L10
Bore Diameter	din	Assumed + Optimized	0.15	m	6.00	in	Optimized for L10
Bearing Width	dw	Assumed + Optimized	0.05	m	2.00	in	None
Rolling Element Diameter	d	Assumed + Optimized	1.27E-02	m	0.50	in	Optimized for L10
Nominal Contact Angle	α	Assumed + Optimized	50.00	deg	50.00	deg	0/5/10/15/20/25/30/35/40/45+
Roller Length	l	=IF(Bhs = "Ball", "N/A", (IF(Bhs = "Roller", "Value Required")))	0.75	m	0.75	in	For roller, length value required!
Ball Diameter Limit (11-28-16 Handout)	dlim	Assumed + Optimized	0.0250	m	0.984	in	None
Calculations (Single Bearing):							
Load-Life Exponent	p	=IF(Bhs="Ball", 3, (IF(Bhs="Roller", 10/3)))	3.33	N/A	3.33	N/A	None
Pitch Diameter	dm	=AVERAGE(dout, din)	0.18	m	7.00	in	None
Table 1.1 Fcm Coefficient	Tc-fcm	=(d* $\cos(\alpha)$)/dm	0.05	N/A	0.0459	N/A	None
Material Coefficient	fcm	= (Table 1.1 - Table 1.6 @Bls @Bhs @i) OR:=IF(AND(Bls="Radial",Bhs="Ball",i = 1), FORECAST(Tc-fcm,OFFSET(TABLES!C11:C50,MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50,1)-1,0,2),OFFSET(TABLES!\$B\$11:\$B\$50,MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50,1)-1,0,2)), IF(AND(Bls="Radial",Bhs="Ball", i = 2), FORECAST(Tc-fcm,OFFSET(TABLES!D11:D50,MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50,1)-1,0,2),OFFSET(TABLES!\$B\$11:\$B\$50,MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50,1)-1,0,2)), IF(AND(Bls="Radial",Bhs="Roller"), FORECAST(Tc-fcm,OFFSET(TABLES!G11:G50,MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50,1)-1,0,2),OFFSET(TABLES!\$B\$11:\$B\$50,MATCH(Tc-fcm, TABLES!\$B\$11:\$B\$50,1)-1,0,2)), IF(AND(Bls="Thrust", Bhs="Ball", a = 45), FORECAST(Tc-fcm,OFFSET(	13668.45	N/A	13668.45	N/A	Integrated: Radial Ball Radial Roller Thrust Ball (45,60,75 ONLY) Thrust Roller (50,65,80 ONLY)

Basic Dynamic Load Capacity	C	=(Eq. 1.5 - 1.11 @Bls @Bhs @d @α) OR: =IF(AND(Bls="Radial", Bhs = "Ball", d<=dlim), (fcm*((i*COS(RADIANS(α)))^0.7)*(Z^(2/3))*(d^1.8))), IF(AND(Bls="Radial", Bhs = "Ball", d>dlim), (fcm*((i*COS(RADIANS(α)))^0.7)*(Z^(2/3))*(d^1.4))), IF(AND(Bls="Radial", Bhs = "Roller"), (fcm*((l*i*COS(RADIANS(α)))^(7/9))*(Z^(3/4))*(d^(29/27)))), IF(AND(Bls="Thrust", Bhs = "Ball", α<>90), (fcm*((i*COS(RADIANS(α)))^0.7)*(TAN(RADIANS(α)))^2*(Z^(2/3))*(d^1.8))), IF(AND(Bls="Thrust", Bhs	254982.81	N	57322.44	lbf	Eq. 1.5-1.11 integrated
Load Ratio	Lr	=IF(Fr = 0, "1000", Fa/Fr)	0.22	N/A	0.22	N/A	If number is larger than 999, assume infinity
Relative Axial Load	RAL	=Fa/(l * Z * d^2)	0.41	N/(mm^2)	59.95	lb/in^2	None
Limit Value	e	=(Table 1.7 - 1.10 @α @Bls @Bhs) OR: use piece-wise linear	1.49	N/A	1.49	N/A	Table 1.7 - 1.10 integrated
Dynamic Radial Load Factor	X	=(Table 1.7 - 1.10 @α @Bls @Bhs) OR: use piece-wise linear	1.79	N/A	1.79	N/A	Table 1.7 - 1.10 integrated
Dynamic Axial Load Factor	Y	=(Table 1.7 - 1.10 @α @Bls @Bhs) OR: use piece-wise linear	0.67	N/A	0.67	N/A	Table 1.7 - 1.10 integrated
Equivalent Load	P	=XFr + YFa	17428.67	N	3918.12	lbf	Eq 1.12
L10 Life (in cycles)	L10(cycles)	=((C/P)^p)*10^6	7.66E+09	Cycles	7.66E+09	Cycles	Eq 1.4
L10 Life (in hours)	L10(hours)	=L10(Cycles) / (Na(RPM)*60min/hr)	35456.96	Hours	35456.96	Hours	None
L10 Life (in days)	L10(days)	=L10(Hours)/24	1477.37	Days	1477.37	Days	None
Design Verifications:							
Life optimization	None	L10(days) > 365	1477.37	Days	1477.37	Days	PASS
Can #of balls/rollers of given diameter fit inside given raceways?	None	(AVERAGE(din, dout)*PI() - (d*Z)^2)/ (AVERAGE(din, dout)*PI()) > 0 circumference of raceway pitch diameter > 2.00 * (ball diameter * # of balls) (NOTE: 2.00 is a safety clearance factor for inner spacing between balls for positioning mechanism)	32%	%	32%	in	PASS

2.1.4.2. General Drawing:



2.1.4.3. Results Summary: A single-direction bearing was not selected for the hydrodynamic thrust bearing because under the *Bearing Manufacturers Association* (pg.24 of L10 Life packet), “when $F_a/F_r < e$ for thrust bearings, it is unsuitable to use single-direction”. A roller was selected over the ball because the L10 life was higher when both calculations were performed. The calculated L10 life for this bearing is 1,477 days. The dimensional recommendations are shown in the drawing above.

3. Problem#2: Hydrostatic Bearing design:

3.1. Problem Statement:

Design a hydrostatic bearing to lift 9,000,000 pounds using water as the lubricant to move a section of the Mile High Stadium.

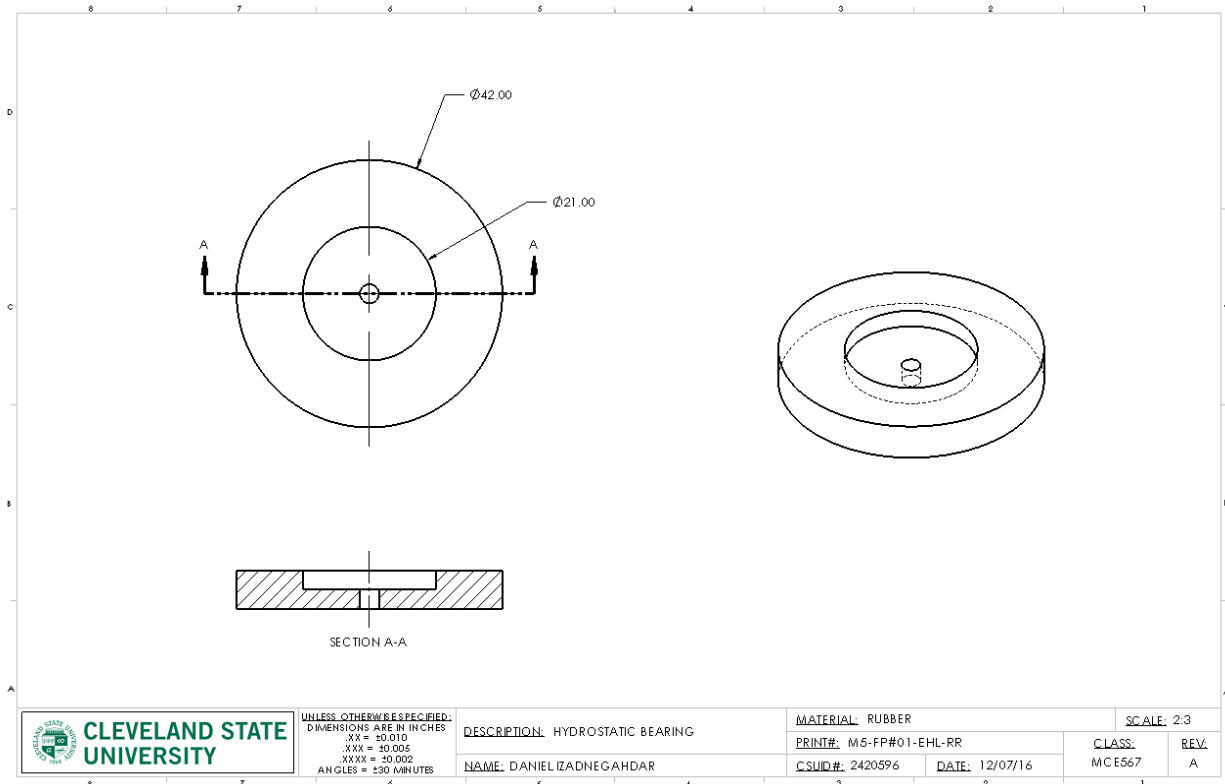
3.2. Background Research:

The current Mile High stadium was designed with rubber circular pads each having a diameter of 4ft using water as the lubricant. The following calculations will estimate values that are reasonable to this application.

3.3. Calculations:

Hydrostatic Bearing Design Calculations							
Description	Symbol	Equation / Condition	SI Units		English Units		Comments
			Value	Unit	Value	Unit	
Given:							
Lubrication Regime	N/A	Given	Hydrostatic	N/A	Hydrostatic	N/A	None
Lubricant	N/A	Given	WATER	N/A	WATER	N/A	None
Absolute viscosity	Eta(n)	Given	8.90E-04	pascal-sec	1.29E-07	psi-sec	None
Lubricant density	Rho(p)	Given	1.00E+03	kg/m^3	0.04	lbf/in^3	None
Total load	Wz(total)	Given	4.00E+07	N	9.00E+06	lbs	None
Assume:							
Pad Type	N/A	Assumed + Optimized	Circular	N/A	Circular	N/A	None
Number of pads	n	Assumed + Optimized	50	N/A	50	N/A	None
Film Thickness	ho	Assumed + Optimized	5.08E-05	m	2.00E-03	in	Adjust to minimize dc and lc
Supply Pressure	Ps	Assumed + Optimized	5.52E+06	Pa	800.00	psi	Adjust for pressure ratio
Pressure ratio	Pr/Ps	Assumed + Optimized	0.30	N/A	0.30	N/A	Adjust for pressure ratio
Radial ratio	ri/ro	Assumed + Optimized	0.50	N/A	0.50	N/A	None
Reynold's number	Re	Assumed + Optimized	400.00	N/A	400.00	N/A	Adjust for dc value
Geometrical Calculations:							
Orifice Discharge Coefficient	Cd	=IF(Re>15 , 0.6, "enter value")	0.600	N/A	0.600	N/A	Must be 0.6 for Re>15
Pad Load	Wz(pad)	=Wz(total)/n	8.01E+05	N	1.80E+05	lbf	None
Load Coefficient	ab	= [@fig13-4@ri/ro@ab line] OR:(1-(ri/ro)^2)/(2*ln(ro/ri))	0.541	N/A	0.541	N/A	None
Bearing pad flow coefficient	qb	= [@fig13-4@ri/ro@qb line] OR: pi/(3*(1-(ri/ro)^2))	1.396	N/A	1.396	N/A	None
Bearing pad power coefficient	Hb	= [@fig13-4@ri/ro@hb line] OR: (2*pi*ln(ro/ri))/(3*[1-(ri/ro)^2]^2)	2.581	N/A	2.581	N/A	None
Recess Pressure	Pr	=(Pr/Ps)*Ps	1.65E+06	Pa	240.000	psi	None
Projected Pad Area	Ap	=wz/(ab * Pr)	0.894	m^2	1386.291	in^2	None
Power Loss	Hp	=Hb*((Wz(pad)/Ap)^2)*((ho^3)/n)	304.67	W	0.409	hp	None
Outer radius	ro	=sqrt(Ap/pi)	0.534	m	21.0	in	None
Inner radius	ri	=ro*(ri/ro)	0.267	m	10.5	in	None
Volumetric Flow Rate	qc	=qb * (wz/Ap) * (ho^3/n)	1.84E-04	m^3/s	11.236	in^3/s	None
Capillary Tube Diameter	dc	=(4*p*qc)/(Re*pi*n)	0.659	m	25.926	in	None
Capillary constant	Kc	=qc*n/(ps-pr)	4.24E-14	N/A	4.24E-14	N/A	None
Capillary Tube Length	lc	=(pi*dc^4)/(128*kc)	1.09E+11	m	4.28E+12	in	None
Volumetric Flow Rate	qo	=qc	1.84E-04	m^3/s	11.236	in^3/s	None
volumetric Flow Rate(gpm)	qo(gpm)	=qo * 15850.3	2.918	gpm	2.918	gpm	qo was converted to gpm to help select a pump. Pumps are typically specked by gpm.
Flow Rate Constant	Ko	=qo / sqrt(Ps - Pr)	9.37E-08	N/A	9.37E-08	N/A	None
Orifice Diameter	do	=sqrt((sqrt(8 * p) * Ko) / (PI * Cd))	0.0021	m	0.083	in	None
Design Verifications:							
Flow Rate Constant	qo(gpm)	qo(gpm) < 3.9 gpm (Affordable mcmaster pumps)	2.918	gpm	2.918	gpm	PASS
Supply Pressure	Ps	Ps < 6.895e+6Pa	5.52E+06	Pa	800.00	psi	PASS
Pressure ratio	Ps - Pr	Ps-Pr >= 3.4E6 Pa	3.86E+06	Pa	560.00	psi	PASS
Capillary Tube Diameter	dc	dc >= 0.6E-4 m	0.659	m	25.926	in	PASS
Reynold's number	Re	Re < 2000	400.00	N/A	400.00	N/A	PASS
l-to-d ratio	lc/dc	lc/dc >= 20	1.65E+11	N/A	1.65E+11	N/A	PASS
Orifice Diameter	do	do >= 5E-4 m	0.002	m	0.083	in	PASS

3.4. General Drawing:



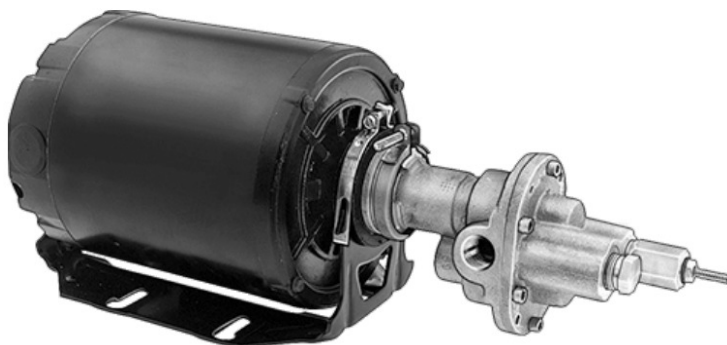
3.5. Recommended Pump:

McMASTER-CARR OVER 555,000 PRODUCTS

(330) 995-5500
 (330) 995-9600 (fax)
 cle.sales@mcmaster.com
 Text 76947

Dispensing Pump
 for Water, 3.9 gpm Maximum Flow

In stock
 \$537.74 Each
 4272K4



Housing Material	Bronze
Priming	Self-Priming
Maximum Flow	3.9 gpm
hp	1/2
Volts AC @ 60	120/240 (1)
Hz (phase)	
Amps	6.8/3.4
Temperature Range	-20° to 210° F
Connections	NPT Female
Pipe Size	3/8
Motor Enclosure Type	Open Dripproof (ODP)

3.6. Results Summary:

For this *hydrostatic* bearing, the estimated pressures are $P_r = 1.65E6$ Pascal, $P_s = 5.52E6$ Pascal, 50 circular pads of $r_o = 0.53m$ and $r_i = 0.27m$ with compensating elements of $d_c = 0.66$ m, $l_c = 1.09E11$ m, and $d_o = 0.0021m$. The calculated flow rate is estimated at 2.92 gpm. A dispensing pump, also known as a positive displacement pump with a flow rate capacity of 3.9gpm was selected for this application. A positive displacement pump was selected to provide a constant fixed flow rate.

4. Problem#3: Bearing Equation Derivation:

4.1. Problem Statement:

Derive an equation for the temperature rise of a thrust bearing in a hydrodynamic lubrication regime.

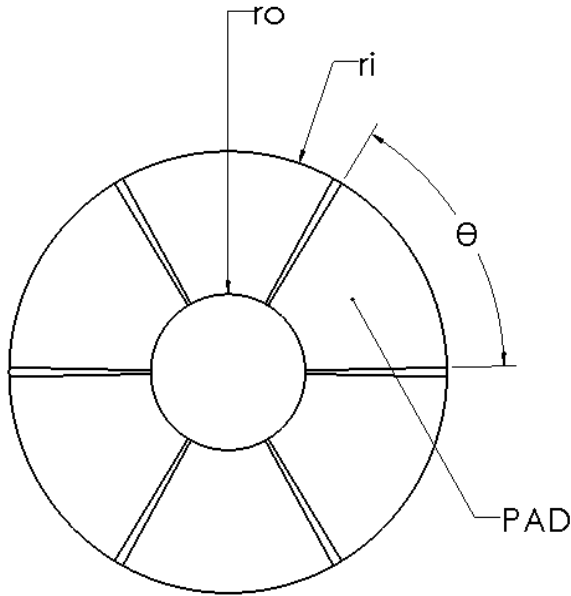
4.2. Approach Summary:

The process for calculating temperature rise for journal bearings in a hydrodynamic lubrication regime was by trial and error using a variety of equations and performance studies. For calculating the same temperature rise in the same regime for a thrust bearing, an equation will be derived by assessing the heat generated from the friction of the bearing pad. A double integral will be done to calculate the area or the total power loss. The geometry of the pad will be assessed, correlated to the friction force generated, and equated to a heat loss that derives a ΔT equation with measureable variables.

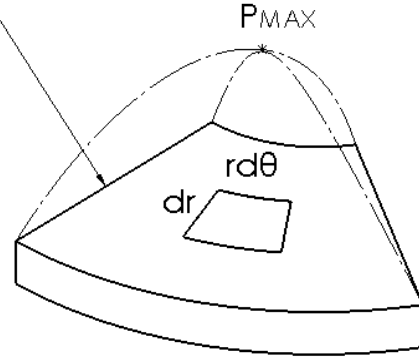
4.3. Symbols , Units, & Values:

- 1) $t_q = \text{Friction Torque (N * m)}$
- 2) $H_{loss} = \text{Power Loss (W)}$
- 3) $H_{loss}^* = \text{Power Loss – Secondary (W)}$
- 4) $w = \text{Angular Velocity } (\frac{rad}{s})$
- 5) $J = \text{Joulean Heat Equivalent} = 9336 \frac{in \cdot lbf}{Btu}$
- 6) $\rho = \text{Density} = 0.0361 \frac{lbf}{in^3}$
- 7) $C_p = \text{Specific Heat} = 0.42 \frac{Btu}{lbm \cdot ^\circ F}$
- 8) $Q_s = \text{Flow Rate} = 5 \frac{in^3}{s}$
- 9) $\Delta T = \text{Temperature Rise}(^\circ F)$
- 10) $f = \text{Friction Force}(lbf)$
- 11) $N = \text{Normal Force}(lbf)$
- 12) $r_o = \text{Outer Radius (m)}$
- 13) $r_i = \text{Inner Radius (m)}$
- 14) $\mu = \text{Coefficient of Friction} = 0.001$
- 15) $\theta = \text{Pad Angle}(^\circ)$
- 16) $\theta^* = \text{Pad Angle(Secondary)} (^\circ)$
- 17) $\partial \text{ or } d = \text{Derivative(not a variable)}$
- 18) $A = \text{Area}(in^2)$
- 19) $P = \text{Pressure} (\frac{lbf}{in^2})$
- 20) $P_{max} = \text{Maximum Pressure} = 450 \frac{lbf}{in^2}$

4.4. Assumptions & Notes:



ZERO PRESSURE ON BOUNDARY



$$r_i \leq r \leq r_o$$

$$0 \leq \theta \leq \theta^*$$

AREA OF ELEMENT $r/\theta = r dr d\theta$

4.5. Equation Derivations:

4.5.1. Solving for t_q :

[EQ4-1]

$$t_q = f * r$$

$$\rightarrow f = \mu * N$$

$$\rightarrow N = P * A$$

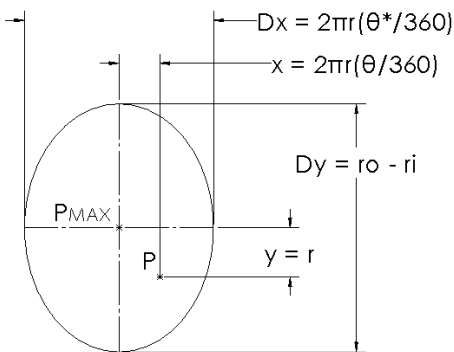
$$t_q = \mu * r * (P * A)$$

$$\partial t_q = \mu * r * (P \partial A)$$

$$\rightarrow \partial A = r \partial r \partial \theta$$

[EQ4-2]

$$t_{q(TOTAL)} = \int_0^{\theta^*} \int_{r_i}^{r_o} \mu * P * r^2 \partial r \partial \theta$$



$$P = P_{max} * \sqrt{1 - \left(\frac{2x}{Dx}\right)^2 - \left(\frac{2y}{Dy}\right)^2}$$

$$P = P_{max} * \sqrt{1 - \left(\frac{2 \left(2\pi r \left(\frac{\theta}{360} \right) \right)}{2\pi r \left(\frac{\theta^*}{360} \right)} \right)^2 - \left(\frac{2r}{(r_o - r_i)} \right)^2}$$

$$t_q = \mu * P_{max} * \int_0^{\theta^*} \int_{r_i}^{r_o} \left(\sqrt{1 - \left(\frac{2 \left(2\pi r \left(\frac{\theta}{360} \right) \right)}{2\pi r \left(\frac{\theta^*}{360} \right)} \right)^2 - \left(\frac{2r}{(r_o - r_i)} \right)^2} \right) * r^2 \partial r \partial \theta$$

Result after double integration and simplification:

$$[EQ4-3] \quad t_q = \left(\frac{1}{6} \right) * \mu * P_{max} * \theta^* * (r_o^3 - r_i^3)$$

4.5.2. Solving for H_{loss} *:

$$[EQ4-4] \quad H_{loss} * = \frac{H_{loss}}{J} \quad \xrightarrow{\quad} \quad H_{loss} = t_q * w$$

$$H_{loss} * = \frac{t_q * w}{J} \quad \left(w = \frac{Na_{RPM}}{60} * 2 * \pi \right)$$

Insert EQ4-3 into EQ4-4:

$$[EQ4-5] \quad H_{loss} * = \frac{\left(\left(\frac{1}{6} \right) * \mu * P_{max} * \theta^* * (r_o^3 - r_i^3) \right) * w}{J}$$

4.5.3. Solving for ΔT (Final Equation):

$$[EQ4-6] \quad H_{loss} * = \rho * C_p * Q_s * \frac{\Delta T}{2}$$

$$[EQ4-7] \quad \Delta T = \frac{2 * H_{loss} *}{\rho * C_p * Q_s}$$

Insert EQ4-5 into EQ4-7:

$$\Delta T = \frac{2 * \left(\frac{\left(\left(\frac{1}{6} \right) * \mu * P_{max} * \theta^* * (r_o^3 - r_i^3) \right) * w}{J} \right)}{\rho * C_p * Q_s}$$

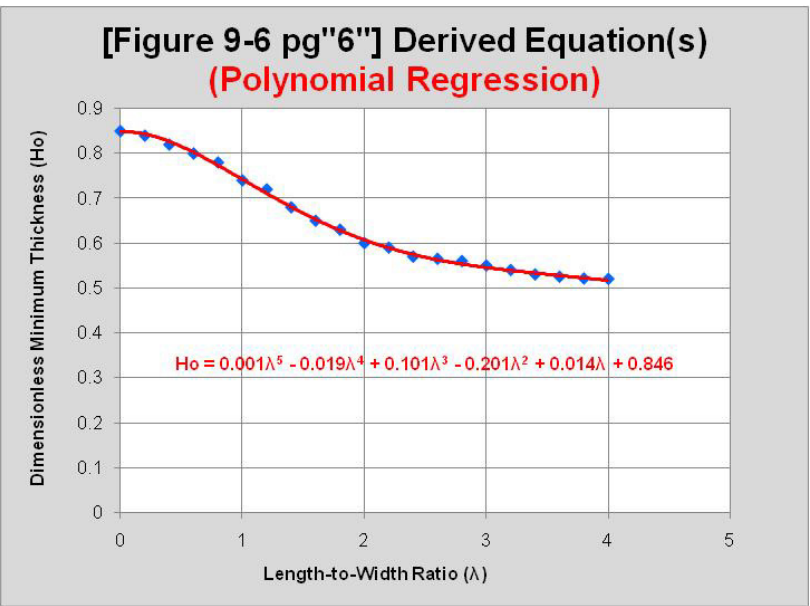
Enter constants from section 4.3:

$$\Delta T = \frac{2 * \left(\frac{\left(\left(\frac{1}{6} \right) * (0.001) * (450) * \theta^* * (r_o^3 - r_i^3) \right) * w}{(9336)} \right)}{(0.0361) * (0.42) * (5)}$$

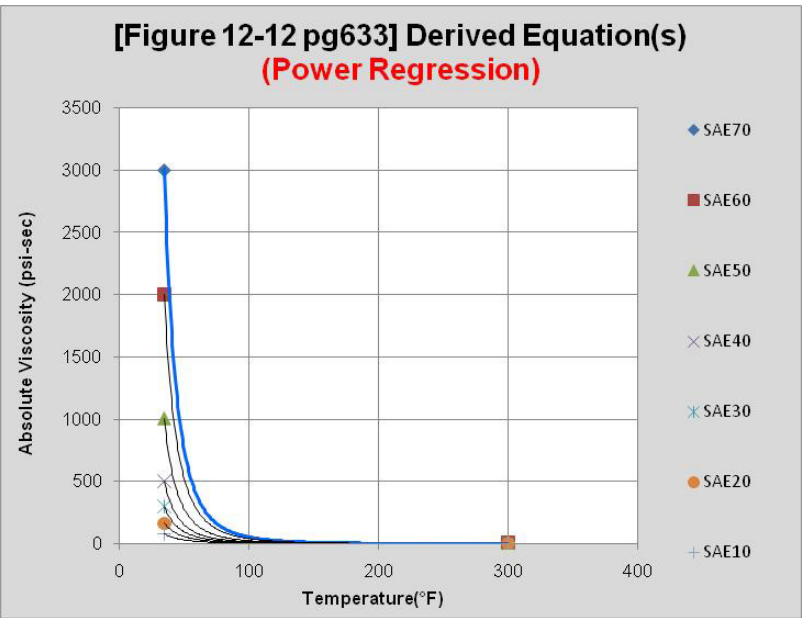
Final result after simplification:

$$[EQ4-8] \Delta T = 0.000212 * r_o^3 * \theta^* * w - 0.000212 * r_i^3 * \theta^* * w$$

5. Appendix:
5.1. Derived Graphs:

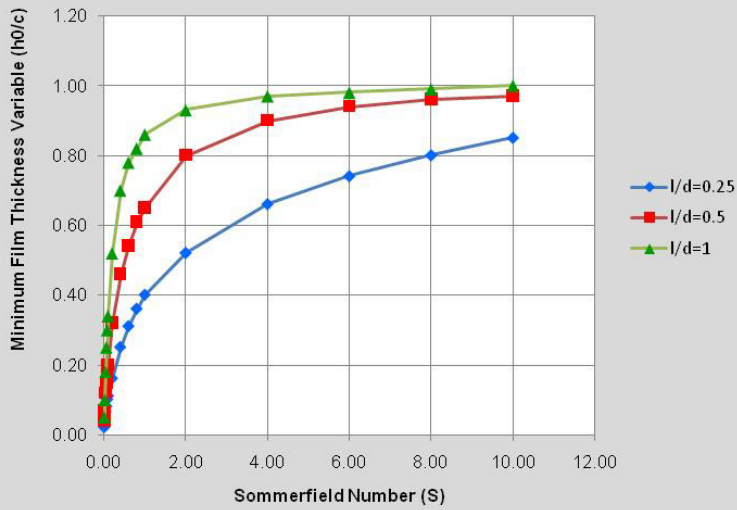


Sampled Data	
λ	H_o
0	0.85
0.2	0.84
0.4	0.82
0.6	0.8
0.8	0.78
1	0.74
1.2	0.72
1.4	0.68
1.6	0.65
1.8	0.63
2	0.6
2.2	0.59
2.4	0.57
2.6	0.565
2.8	0.56
3	0.55
3.2	0.54
3.4	0.53
3.6	0.525
3.8	0.521
4	0.52



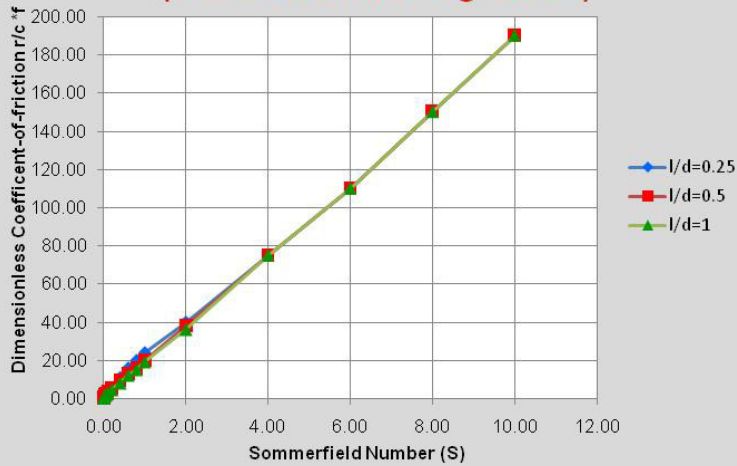
X	Y	Lube	FACTOR	EXP
35	3000	SAE70	2.00E+09	-3.8
300	0.85			
35	2000	SAE60	9.00E+08	-3.67
300	0.75			
35	1000	SAE50	3.00E+08	-3.52
300	0.52			
35	500	SAE40	6.00E+07	-3.28
300	0.43			
35	300	SAE30	2.00E+07	-3.11
300	0.375			
35	160	SAE20	4.90E+06	-2.91
300	0.310			
35	75	SAE10	88338	-2.63
300	0.26			

[Figure 12-16 pg"636"] Derived Equation(s)
(Piecewise Linear Regression)



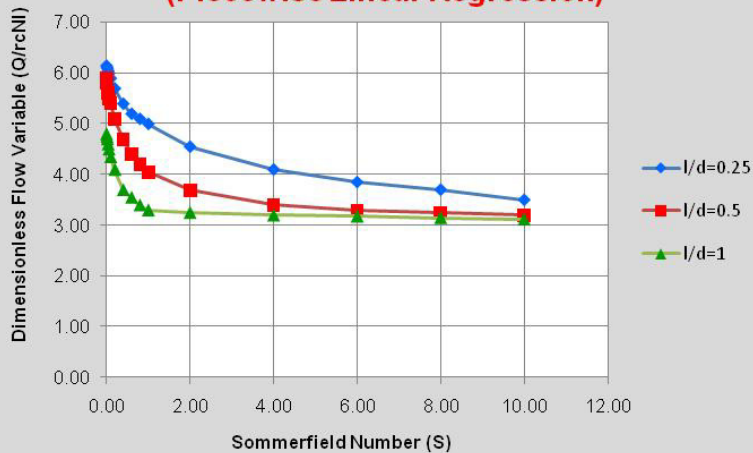
Sampled Data			
S	l/d=0.25	l/d=0.5	l/d=1
0.00	0.00	0.00	0.00
0.01	0.02	0.04	0.05
0.02	0.04	0.07	0.10
0.04	0.07	0.12	0.18
0.06	0.08	0.15	0.25
0.08	0.10	0.18	0.30
0.10	0.11	0.20	0.34
0.20	0.16	0.32	0.52
0.40	0.25	0.46	0.70
0.60	0.31	0.54	0.78
0.80	0.36	0.61	0.82
1.00	0.40	0.65	0.86
2.00	0.52	0.80	0.93
4.00	0.66	0.90	0.97
6.00	0.74	0.94	0.98
8.00	0.80	0.96	0.99
10.00	0.85	0.97	1.00

[Figure 12-18 pg"639"] Derived Equation(s)
(Piecewise Linear Regression)



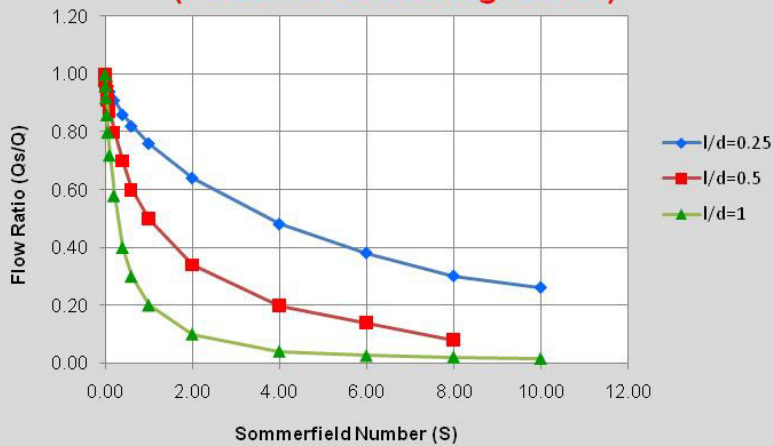
Sampled Data			
S	l/d=0.25	l/d=0.5	l/d=1
0.00	0.00	0.00	0.00
0.01	0.90	0.80	0.80
0.02	1.40	1.30	1.10
0.04	2.20	1.90	1.60
0.06	2.40	2.40	1.90
0.10	4.25	3.40	2.60
0.20	5.40	5.50	4.50
0.40	11.00	9.50	8.00
0.60	16.00	13.00	12.00
0.80	20.00	16.00	15.00
1.00	24.00	20.00	19.00
2.00	40.00	38.00	36.00
4.00	75.00	75.00	75.00
6.00	110.00	110.00	110.00
8.00	150.00	150.00	150.00
10.00	190.00	190.00	190.00

[Figure 12-19 pg"640"] Derived Equation(s)
(Piecewise Linear Regression)



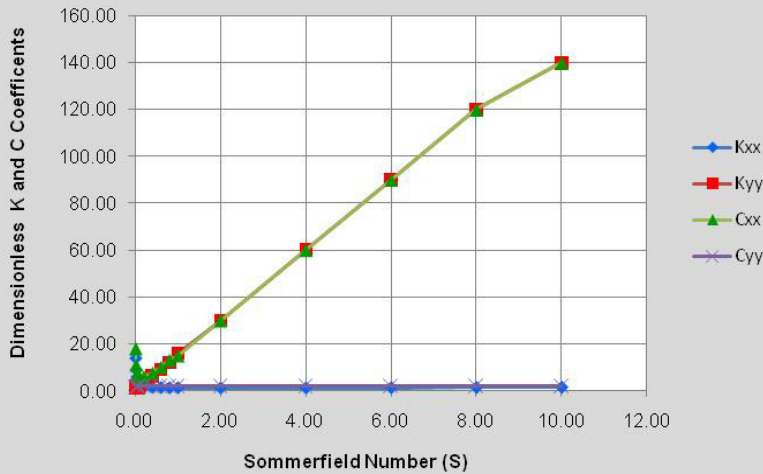
Sampled Data			
S	l/d=0.25	l/d=0.5	l/d=1
0.00	6.15	5.90	4.80
0.01	6.12	5.80	4.75
0.02	6.10	5.70	4.70
0.04	6.05	5.60	4.60
0.06	6.00	5.50	4.50
0.10	5.90	5.40	4.35
0.20	5.70	5.10	4.10
0.40	5.40	4.70	3.70
0.60	5.20	4.40	3.55
0.80	5.10	4.20	3.40
1.00	5.00	4.05	3.30
2.00	4.55	3.70	3.25
4.00	4.10	3.40	3.20
6.00	3.85	3.30	3.18
8.00	3.70	3.25	3.15
10.00	3.50	3.20	3.12

**[Figure 12-20 pg"640"] Derived Equation(s)
(Piecewise Linear Regression)**



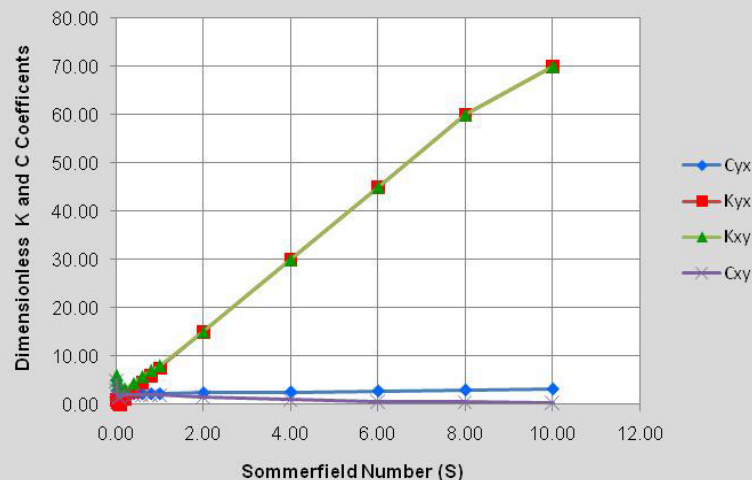
Sampled Data			
S	$l/d=0.25$	$l/d=0.5$	$l/d=1$
0.00	1.00	1.00	1.00
0.01	0.99	0.98	0.96
0.02	0.98	0.96	0.92
0.04	0.98	0.94	0.86
0.06	0.96	0.91	0.80
0.10	0.94	0.87	0.72
0.20	0.91	0.80	0.58
0.40	0.86	0.70	0.40
0.60	0.82	0.60	0.30
1.00	0.76	0.50	0.20
2.00	0.64	0.34	0.10
4.00	0.48	0.20	0.04
6.00	0.38	0.14	0.03
8.00	0.30	0.08	0.02
10.00	0.26	0.05	0.02

**[Figure B-3 pg"76"] Derived Equation(s)
(Piecewise Linear Regression)**



Sampled Data (for $L/D = 1$ ONLY)				
S	K_{xx}	K_{yy}	C_{xx}	C_{yy}
0.01	14.00	2.00	18.00	1.6
0.02	9.00	1.40	11.00	1.8
0.04	5.80	1.25	7.00	1.9
0.06	4.20	1.40	6.00	2
0.08	3.40	1.60	5.80	2
0.10	3.00	2.10	5.70	2
0.20	1.70	4.00	6.00	2.1
0.40	1.40	6.25	7.50	2.15
0.60	1.30	9.00	10.00	2.2
0.80	1.20	12.00	13.00	2.2
1.00	1.20	16.00	15.00	2.2
2.00	1.05	30.00	30.00	2.2
4.00	1.10	60.00	60.00	2.2
6.00	1.20	90.00	90.00	2.2
8.00	1.30	120.00	120.00	2.2
10.00	1.40	140.00	140.00	2.2

**[Figure B-4 pg"77"] Derived Equation(s)
(Piecewise Linear Regression)**



Sampled Data (for $L/D = 1$ ONLY)				
S	C_{yx}	K_{yx}	K_{xy}	C_{xy}
0.01	5.00	1.10	6.00	5.00
0.02	2.80	0.95	5.00	3.00
0.04	1.80	0.55	4.00	2.00
0.06	1.60	0.28	3.70	1.70
0.08	1.60	0.10	3.65	1.60
0.10	1.60	0.00	3.60	1.60
0.20	2.20	1.30	3.20	2.00
0.40	2.20	2.50	4.30	2.10
0.60	2.20	4.50	5.75	2.00
0.80	2.20	6.00	7.00	2.00
1.00	2.20	7.50	8.00	2.00
2.00	2.40	15.00	15.00	1.60
4.00	2.60	30.00	30.00	1.00
6.00	2.80	45.00	45.00	0.70
8.00	3.00	60.00	60.00	0.53
10.00	3.20	70.00	70.00	0.40

Table 1.1

Conditions		
Bearing Hardware Style	Ball	Ball
Bearing Hardware Style	Radial	Radial
Special	None	Groove
Angle Lower Limit	N/A	N/A
Angle Upper Limit	N/A	N/A
# of Rows of Rolling Element	1.00	2.00
Fcm Coefficient	Fcm Values	
0.01	2875	2717
0.02	3537	3349
0.03	3982	3774
0.04	4327	4100
0.05	4614	4367
0.06	4851	4594
0.07	5049	4782
0.08	5217	4940
0.09	5365	5078
0.10	5483	5197
0.11	5592	5296
0.12	5681	5385
0.13	5750	5454
0.14	5809	5503
0.15	5859	5543
0.16	5888	5582
0.17	5908	5602
0.18	5918	5612
0.19	5928	5612
0.20	5918	5612
0.21	5908	5592
0.22	5888	5582
0.23	5859	5553
0.24	5829	5523
0.25	5790	5510
0.26	5750	5444
0.27	5701	5394
0.28	5641	5345
0.29	5592	5296
0.30	5528	5236
0.31	5464	5177
0.32	5394	5118
0.33	5325	5049
0.34	5256	4980
0.35	5177	4919
0.36	5108	4831
0.37	5029	4762
0.38	4940	4683
0.39	4861	4604
0.40	4782	4525

Note: Numbers OCR Generated from T

Table 1.2

Conditions	
Bearing Hardware Style	Roller
Bearing Hardware Style	N/A
Special	Cylindrical
Angle Lower Limit	N/A
Angle Upper Limit	N/A
# of Rows of Rolling Element	N/A
Fcm Coefficient	Fcm Values
0.010	5149
0.020	6009
0.030	6573
0.040	6987
0.050	7324
0.060	7600
0.070	7828
0.080	8016
0.090	8184
0.100	8322
0.110	8431
0.120	8539
0.130	8609
0.140	8668
0.150	8718
0.160	8747
0.170	8767
0.180	8778
0.190	8778
0.200	8767
0.210	8747
0.220	8718
0.230	8678
0.240	8648
0.250	8589
0.260	8539
0.270	8480
0.280	8452
0.290	8342
0.300	8273
0.310	8204
0.320	8125
0.330	8036
0.340	7946
0.350	7857
0.360	7768
0.370	7669
0.380	7580
0.390	7482
0.400	7373
0.410	7274
0.420	7165
0.430	7057
0.440	6948
0.450	6840
0.460	6731
0.470	6622
0.480	6503
0.490	6383
0.500	6276

Note: Numbers OCR Ge

Table 1.7

Conditions										
Bearing Hardware Style	Ball									
Bearing Hardware Style	Radial									
Special	None									
Angle Lower Limit	0									
Angle Upper Limit	45									
# of Rows of Rolling Element			1				2			
α	$F_a/Zd^{1/2}$	e	X	Y	X	Y	X	Y	X	Y
0	25	0.19	1	0	0.56	2.3	1	0	0.56	2.3
0	50	0.22				1.99				1.99
0	100	0.26				1.71				1.71
0	150	0.28				1.55				1.55
0	200	0.3				1.45				1.45
0	300	0.34	1	0	0.56	1.31	1	0.78	0.78	1.31
0	500	0.38				1.15				1.15
0	750	0.42				1.04				1.04
0	1000	0.44				1				1
5	25	0.23				2.3				3.74
5	50	0.26	1	0	0.56	1.99	1	0.78	0.78	3.23
5	100	0.3				1.71				2.78
5	150	0.34				1.55				2.52
5	200	0.36				1.45				2.36
5	300	0.4				1.31				2.13
5	500	0.45	1	0	0.46	1.15	1	0.75	0.75	1.87
5	750	0.5				1.04				1.69
5	1000	0.52				1				1.63
10	25	0.29				1.88				3.06
10	50	0.32				1.71				2.78
10	100	0.36	1	0	0.44	1.52	1	0.72	0.72	2.47
10	150	0.38				1.41				2.29
10	200	0.4				1.34				2.18
10	300	0.44				1.23				2
10	500	0.49				1.1				1.79
10	750	0.54	1	0	0.44	1.01	1	0.72	0.72	1.64
10	1000	0.54				1				1.63
15	25	0.38				1.47				2.39
15	50	0.4				1.4				2.28
15	100	0.43				1.3				2.11
15	150	0.46	1	0	0.44	1.23	1	0.72	0.72	2
15	200	0.47				1.19				1.93
15	300	0.5				1.12				1.82
15	500	0.56				1.02				1.56
15	750	0.56				1				1.63
15	1000	0.56	1	0	0.44	1	1	0.72	0.72	1.63
20	0	0.57				0.43				1.09
25	0	0.68				0.41				0.92
30	0	0.8				0.39				0.78
35	0	0.95				0.37				0.66
40	0	1.14	1	0	0.44	0.35	1	0.72	0.72	0.55
45	0	1.24				0.33				0.47

Table 1.12

L Rating	Reliability %	Life Factor
0.06	99.94	0.053
0.10	99.9	0.053
0.20	99.8	0.08
0.50	99.5	0.15
1.00	99	0.21
2.00	98	0.33
3.00	97	0.44
4.00	96	0.53
5.00	95	0.62
10.00	90	1